

Rock Jobs: Which Property Matters?

BUILD • Science • 40 minutes

I can identify which tested or observed rock property matters for a stated everyday job.

Prepare before pupils arrive

Setup: Prepare four job briefs and separate evidence icons for scratch resistance, water passage, shape and appearance. Do not present a winning sample before the property question.

Safety: All scenarios are classroom design models, not construction approval. Use images/director routes whenever touch, weight or sensory demand creates a barrier.

Equipment: Four supplied job briefs, W10 scratch and W11 water evidence icons, property-question grid and pencil or communication aid.

Safety: Use cards or pre-screened lightweight samples on trays. No real construction, site access, heavy lifting, scratching, wetting, breaking, tasting, smelling or collecting is required.

Fallback: Use one large job picture at a time and fixed property choices. An adult scribes the pupil's exact because link.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Identify everyday uses of different rocks (links D&T / Construction).

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 4 min	Last lesson: what did the water-passage result measure? Last lesson: why did a lower value not mean waterproof forever?	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.
I do · 5 min	property — what the material is like purpose — the job it must do suitable — works for this job Read the brief before the samples. A rain-facing cover, hard-wearing path, indoor display and drainage layer do different jobs.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.

Stage / total time	Teaching move	Adult support / response
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	A job can need more than one property. A path may need resistance to surface wear and a suitable walking surface; a display may value pattern while still needing stability.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Route each brief clue to WATER CONTROL, WEAR / SURFACE, SHAPE / APPEARANCE or NEEDS ANOTHER CHECK. Read the job clue.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Create property-question cards for four everyday rock jobs before any candidate sample is chosen. Record: Four job-property questions, two because explanations, one needs-another-check decision and one model limit.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Explain which property matters for one job and why another property does not answer it. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move; written closure is optional when Voice and Audience are observable.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: Which prepared sample lets less water through and can form overlapping flat pieces?

Correct. Water passage and useful flat shape both connect directly to the stated roof-cover job.

Misconception repair

Repaired. Suitable is always suitable for this stated purpose, using relevant evidence.

Guided checkpoint

Correct. Water passage and useful flat shape both connect directly to the stated roof-cover job. Choose the property evidence that helps the cover deal with rain and overlap across the model roof.

Rain-cover clue

WATER CONTROL. Water control matters: use the sealed water-passage evidence. Which tested property directly follows the route of rain?

Drainage clue

WATER CONTROL. Water control matters here too, but the useful direction is allowing drainage rather than blocking passage. This job is about where water should travel.

Busy-path clue

WEAR / SURFACE. Wear evidence matters: relative scratch resistance can inform the supplied-sample comparison. Which property test was about marks and surface wear?

Walking-surface clue

WEAR / SURFACE. Surface evidence matters, while the card honestly keeps the full safety decision outside the classroom model. This clue describes the surface people contact.

Overlap clue

SHAPE / APPEARANCE. Shape evidence matters because the pieces must overlap across the model. Which target is about the form or visible structure of the pieces?

Display clue

SHAPE / APPEARANCE. Appearance is relevant because the purpose explicitly includes a visual display. For this job, appearance is named in the brief rather than used as a substitute for test data.

Dark-colour claim

NEEDS ANOTHER CHECK. Needs another check: colour does not supply the three different property results claimed. Does one colour observation answer strength, scratch resistance and permeability?

Hardest-is-best claim

NEEDS ANOTHER CHECK. Needs another check: one property cannot settle every purpose. Compare what a path needs with what a rain cover needs.

Model limitation

The briefs simplify real design. Cost, thickness, treatment, strength, installation, safety and local standards also matter; classroom evidence cannot approve real construction.

Independent evidence

Four job-property questions, two because explanations, one needs-another-check decision and one model limit.

Supported: Match one key property picture to three clear jobs and complete one because sentence. Use: For a ___, we need to know ___ because ___.

Standard: Complete all four briefs, naming the relevant property evidence and one missing check for each. Keep separate columns for job, property question and evidence needed.

Stretch: Explain a two-property trade-off and why opposite permeability behaviour can suit different water jobs. Include one real-design factor the classroom brief does not model.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: [SCI_B_W12A_Rock_Jobs_Which_Property_Matters_Explore.html](#)

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.